

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech. Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester : III Subject Code & Name: Electronics Devices and circuits(BTECP302)

Max Marks: 60

Date:07/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks		
Q. 1	Objective type questions. (Compulsory Question)		12		
1	A Field Effect transistor (FET) operates on	CO1	1		
	<table border="1"> <tr> <td>a. Minority carriers only</td> <td>b. Majority carriers only</td> <td>c. Both majority & minority carriers</td> <td>d. None of these</td> </tr> </table>			a. Minority carriers only	b. Majority carriers only
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2	At Pinch off voltage drain current for FET is	CO1	1		
	<table border="1"> <tr> <td>a. Infinite</td> <td>b. Zero</td> <td>c. Both a & b</td> <td>d. None of these</td> </tr> </table>			a. Infinite	b. Zero
a. Infinite	b. Zero	c. Both a & b	d. None of these		
3	The percentage voltage regulation for voltage supply providing 100v with no load and 95v at full load is	CO5	1		
	<table border="1"> <tr> <td>a.0.53%</td> <td>b. 5.26%</td> <td>c. 0.05%</td> <td>d. None of these</td> </tr> </table>			a.0.53%	b. 5.26%
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4	Which of the following circuit is known as emitter follower	CO5	1		
	<table border="1"> <tr> <td>a. Zener shunt regulator</td> <td>b. Transistor shunt regulator</td> <td>c. Transistor series regulator</td> <td>d. Both a & b</td> </tr> </table>			a. Zener shunt regulator	b. Transistor shunt regulator
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5	Two stages of cascaded amplifier have gains of 50 & 20. Then overall voltage gain is	CO3	1		
	<table border="1"> <tr> <td>a.1000</td> <td>b. 30</td> <td>c. 70</td> <td>d. 300</td> </tr> </table>			a.1000	b. 30
a.1000	b. 30	c. 70	d. 300		
6	For impedance matching we can prefer	CO3	1		

	a. RC coupled amplifier	b. Direct coupled amplifier	c. Transformer coupled amplifier	d. None of these		
7	An ideal opamp has				CO2	1
	a. Infinite A_v	b. Zero R_o	c. Infinite R_i	d. All the above		
8	Crystal oscillators are preferred than LC oscillators because of				CO3	1
	a. High Q-factor	b. High frequency stability	c. Small size of crystal	d. Both a & b		
9	The voltage gain of voltage follower is				CO2	1
	a. Less than unity	b. Equal to unity	c. Greater than unity	d. other than unity		
10	The drawback of negative feedback is				CO3	1
	a. Reduced gain	b. Frequency dependence	c. Temperature instability	d. None of these		
11	E-MOSFET is an				CO1	1
	a. Normally ON MOSFET	b. Normally OFF MOSFET	c. Normally ON/OFF MOSFET	d. None of these		
12	Which of the following is passive transducer				CO4	1
	a. Thermocouple	b. Piezoelectric transducer	c. Thermistor	d. photovoltaic cell		
Q. 2	Solve the following.					12
A)	Explain working of N-channel JFET with neat diagram.				CO1	6
B)	Draw and explain drain characteristics of E-MOSFET.				CO1	6
Q. 3	Solve the following.					12
A)	Draw and explain in detail the block diagram of Opamp.				CO2	6
B)	Explain Opamp as Non-inverting amplifier with suitable circuit diagram & waveforms.				CO2	6
Q. 4	Solve Any Two of the following.					12

A)	Explain construction and working of Hartley oscillator.	CO3	6
B)	Draw and explain RC-coupled amplifier.	CO3	6
C)	List different feedback topologies. Explain voltage series feedback connection with the help of block diagram.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain three terminal positive voltage regulators.	CO5	6
B)	Explain the operation of transistor shunt regulator.	CO5	6
C)	Draw pin diagram of IC 723 and list its various features.	CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain construction and working of LVDT.	CO4	6
B)	Draw construction and explain working of LDR.	CO4	6
C)	Explain construction and working of Thermocouple.	CO4	6
	*** End ***		